FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

id0114q2800600

Warning

- **Fuel is highly flammable and dangerous. Fuel line spills and leakage can cause serious injury or death, and damage to equipment. If fuel comes into contact with the skin and eyes it will cause irritation. To prevent this, always perform the "Fuel Line Safety Procedure" referring to the "BEFORE REPAIR PROCEDURE".**
- **A person charged with static electricity could cause a fire or explosion, resulting in death or serious injury. Before servicing the fuel system, make sure to discharge static electricity by touching a vehicle.**

Caution

- **If parts are exposed to the fuel, their function may be decreased or a fire may occur. Spread a rag to absorb any leaked fuel.**
- **Do not let foreign matter get into the injection pipe to prevent clogging. In addition, always install the injection pipe to the specified tightening torque.**

Note

- If the fuel injector is replaced, perform the "Operation After Replacing Fuel Injector".

Operation After Replacing Fuel Injector

1. If the fuel injector is replaced, perform the following procedure.

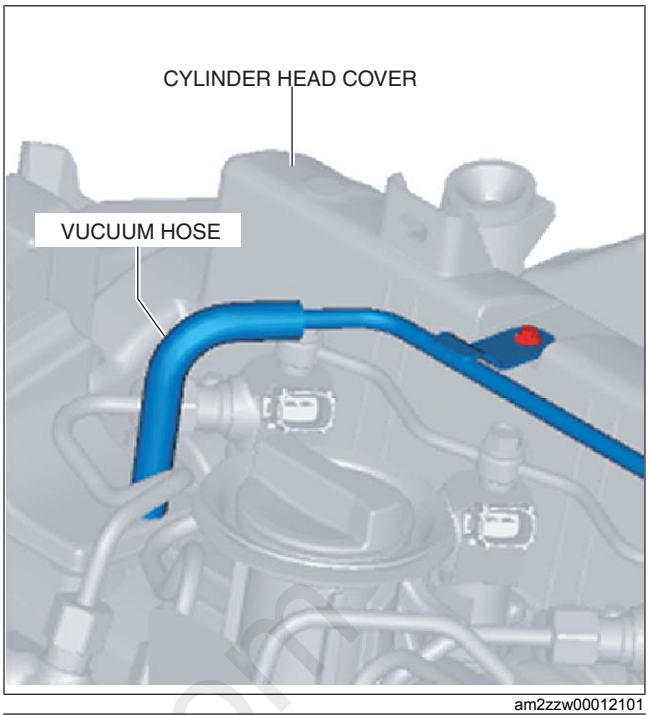
STEP	ACTION	PAGE/CONDITION
1	Perform fuel injector code programming.	(See FUEL INJECTOR CODE PROGRAM [SKYACTIV-D 1.5].)
2	Perform fuel injector data reset procedure.	(See FUEL INJECTOR DATA RESET [SKYACTIV-D 1.5].)
3	Clear the DTCs.	(See OBD READ/CLEAR DIAGNOSTIC TEST RESULTS [SKYACTIV-D 1.5].)
4	Switch the ignition off.	—
5	Wait for 20 s or more.	—
6	Perform KOEO self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
7	Perform KOER self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
8	Perform fuel injector injection amount correction.	(See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].)
9	Clear the DTCs.	(See OBD READ/CLEAR DIAGNOSTIC TEST RESULTS [SKYACTIV-D 1.5].)

Fuel injector Removal/Installation

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Complete the "BEFORE SERVICE PRECAUTION". (See BEFORE SERVICE PRECAUTION [SKYACTIV-D 1.5].)

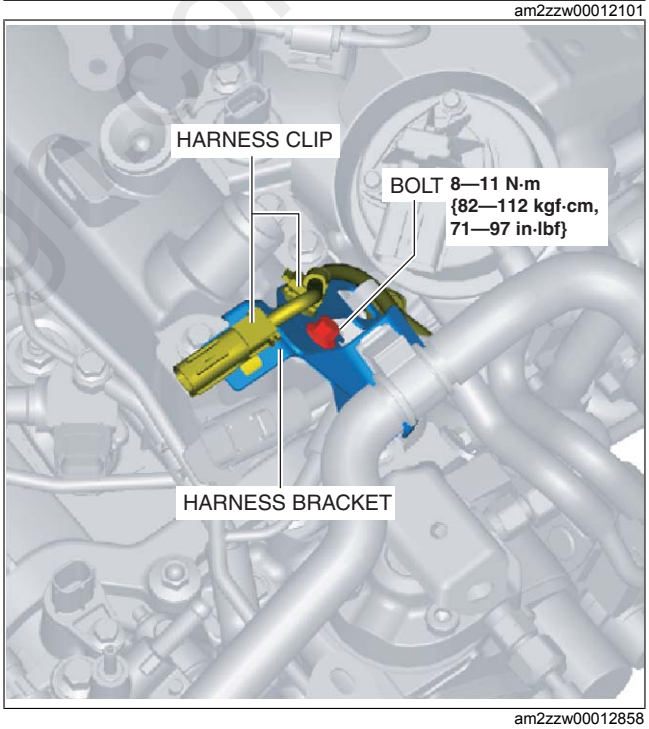
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4. Disconnect the vacuum hose connected to the cylinder head cover.



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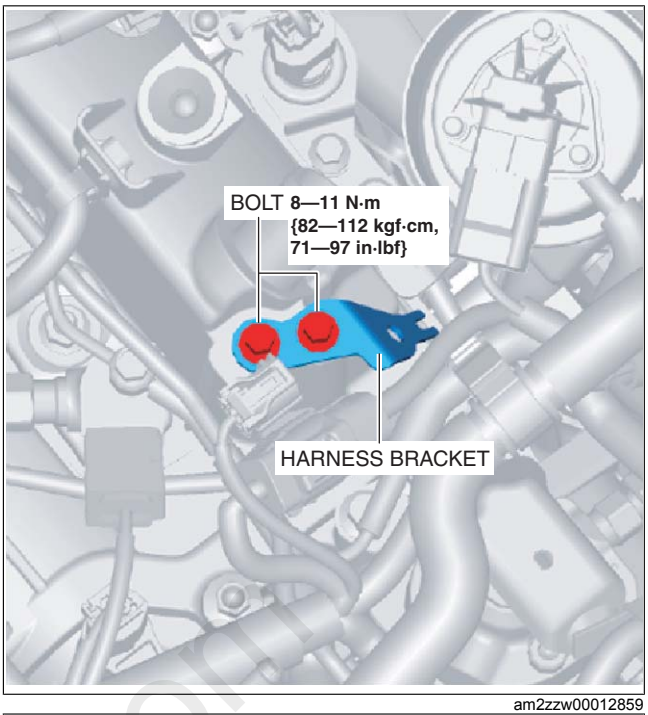
5. Remove the wiring harness clips and bolt shown in the figure and set the wiring harness bracket aside so that it does not interfere with the servicing.



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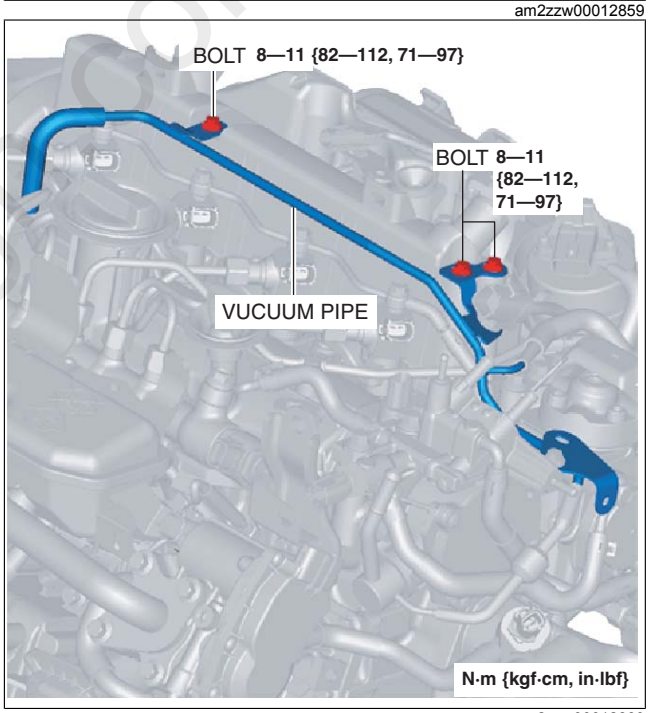
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6. Remove the bolts, and remove the wiring harness bracket.



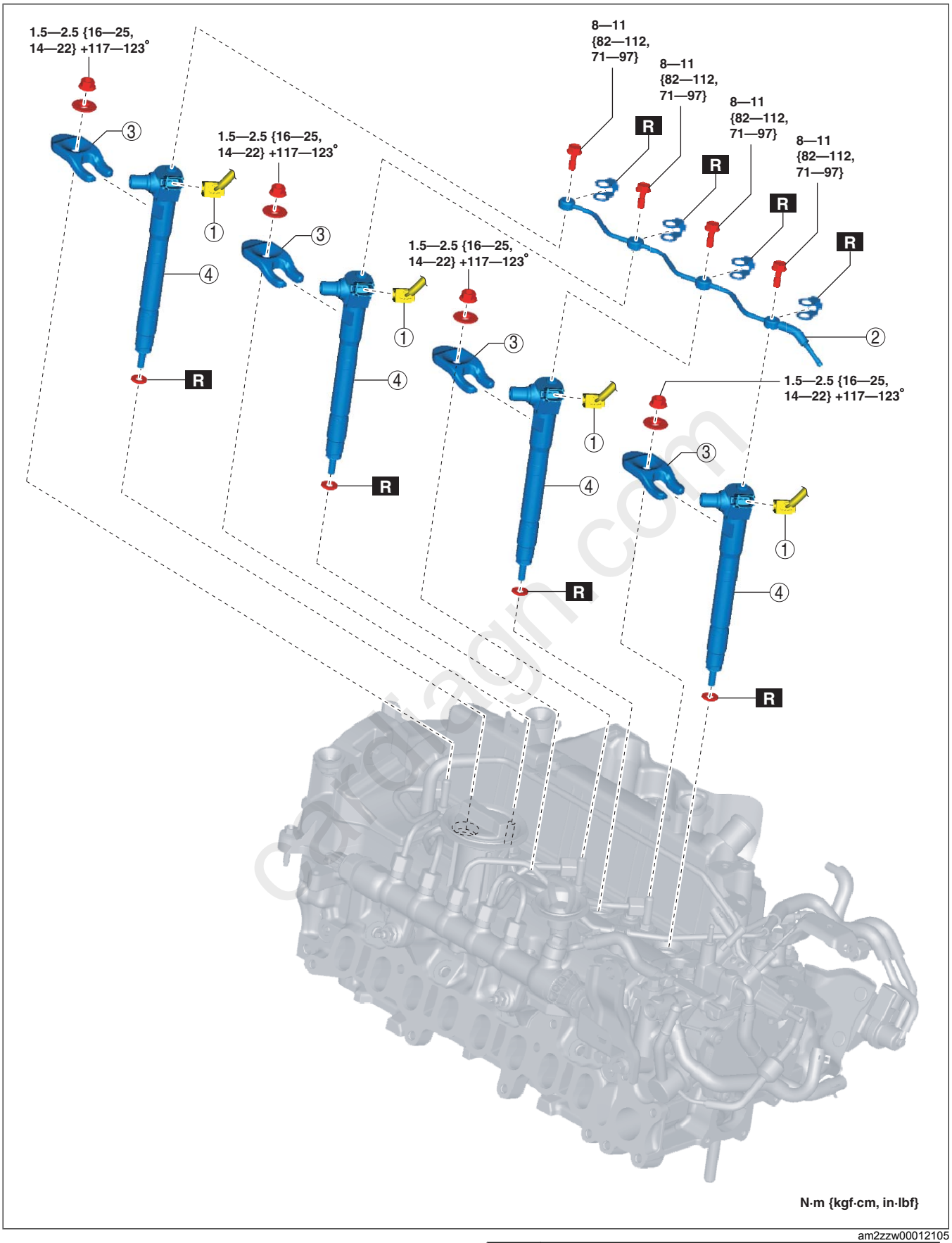
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7. Remove the vacuum pipe installation bolt and set the vacuum pipe aside so that it does not interfere with the servicing.
8. Remove the injection pipe (fuel injector side). (See INJECTION PIPE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
9. Remove in the order shown in the figure.
10. Install in the reverse order of removal.
11. Complete the “AFTER SERVICE PRECAUTION”. (See AFTER SERVICE PRECAUTION [SKYACTIV-D 1.5].)



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1	Fuel injector connector
2	Fuel return pipe
3	Fuel injector bracket
4	Fuel injector (See Operation After Replacing Fuel Injector.) (See Fuel injector installation note.)

Fuel injector installation note

1. Remove the injection pipe (supply pump side). (See INJECTION PIPE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Loosen the common rail.
3. Tighten the bolts shown in the figure to the specified torque.

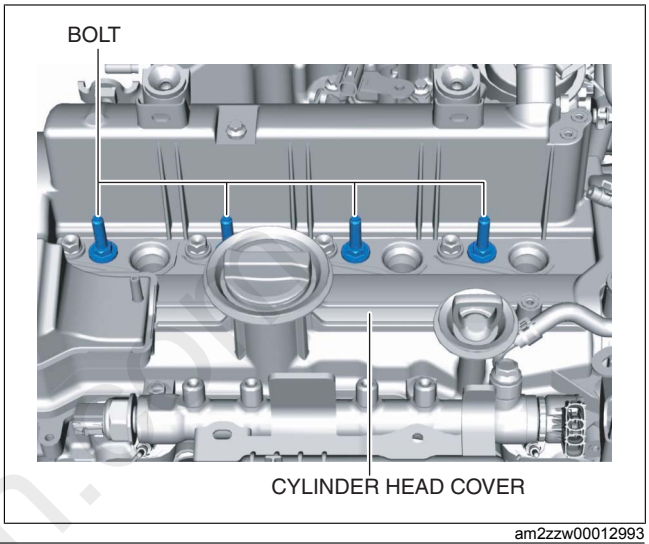
Caution

- **The bolts shown in the figure may loosen when removing the fuel injector bracket because the nuts assembled to the bolts are removed. Tighten the bolts shown in the figure to the specified torque before installing the fuel injector bracket.**

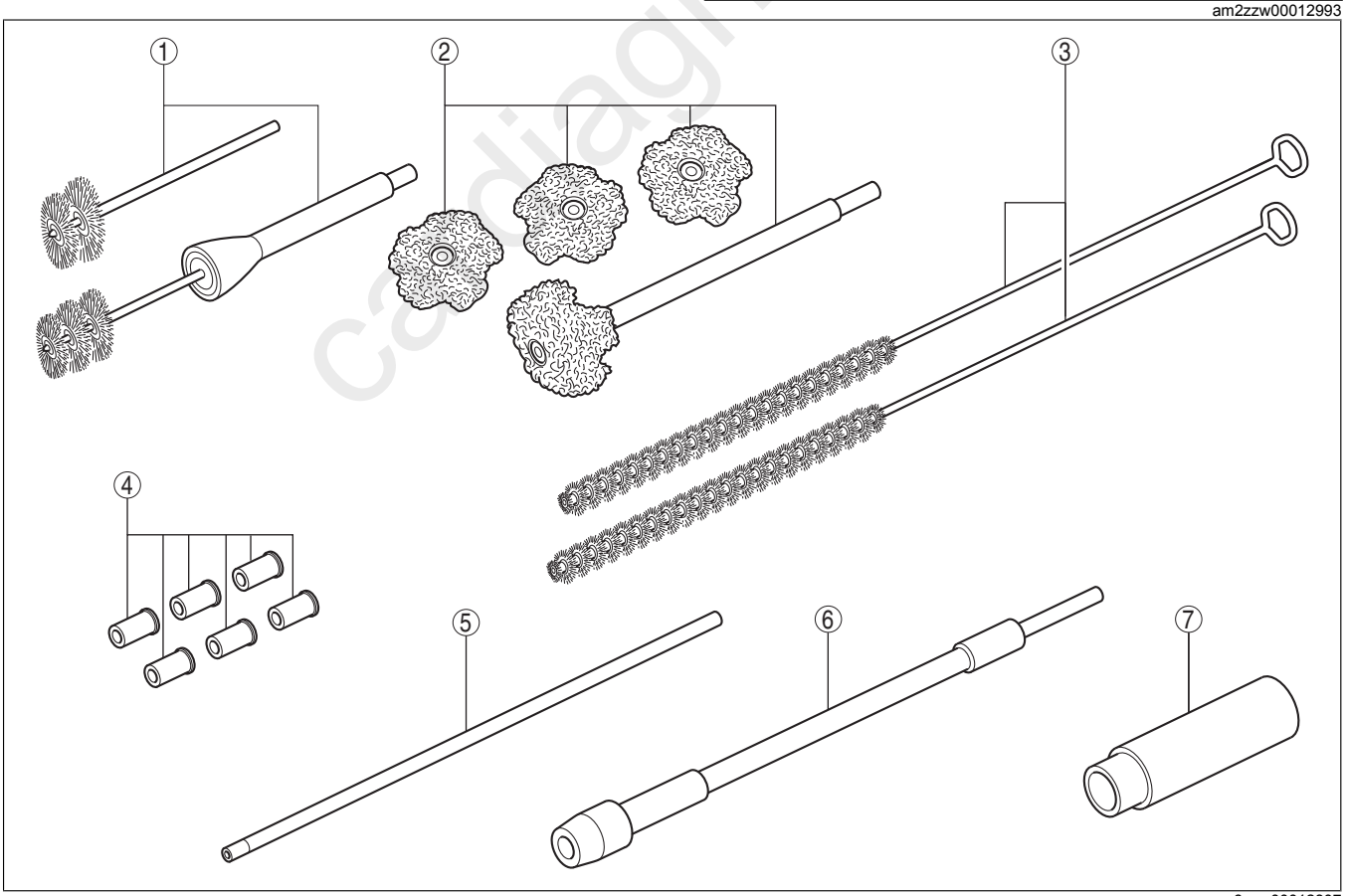
Tightening torque
15—20 N·m {1.6—2.0 kgf·m, 12—14 ft·lbf}

Note

- Clean the injector hole and the bore using the following procedure and the **SSTs (49 0000 203)** shown in the figure.



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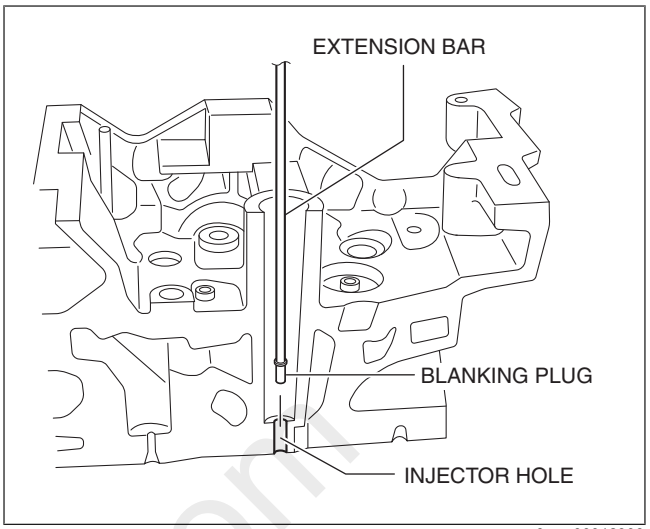
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1	Round wire brush
2	Fleece brush
3	Wire brush
4	Blanking plug

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5	Extension bar
6	Flexible shaft
7	Holder

4. Install the blanking plug to the end of the extension bar.
5. Insert the blanking plug into the injector hole, rotate the extension bar counterclockwise, and then block the injector hole using the blanking plug.



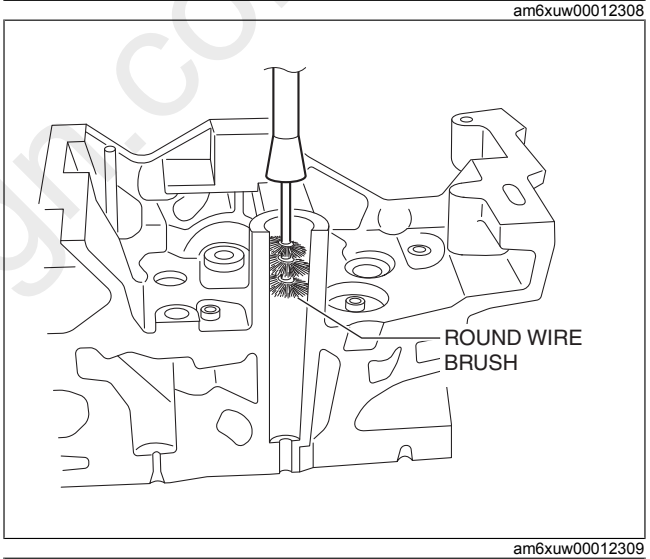
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6. Install the round wire brush to the end of the flexible shaft.

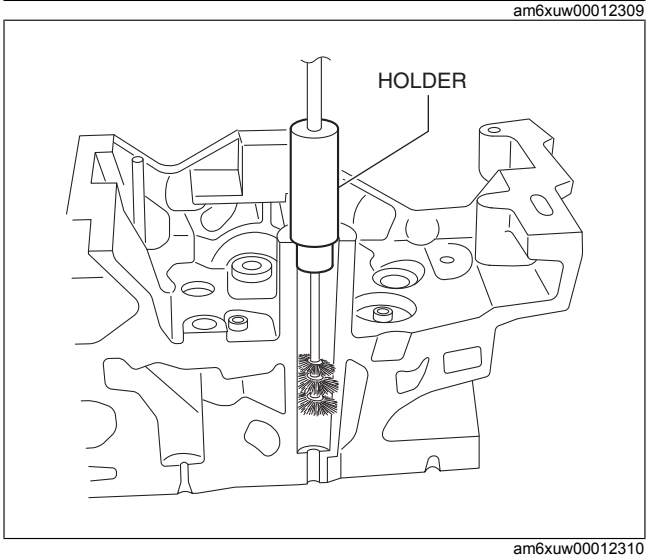
Caution
- **Install the holder and clean the bore to prevent damage to the bore.**

7. Connect the flexible shaft installed to the round wire brush to the drill, and clean the bore.
8. Clean the carbon that was cut off using an air gun.
9. Cut (reference: diameter approx. 23 mm) the fleece brush so that it reaches the bottom surface, and install it to the flexible shaft.

Caution
- **Clean the bottom surface while verifying the cleaning surface condition. If the bottom surface is cleaned excessively, it could be damaged.**



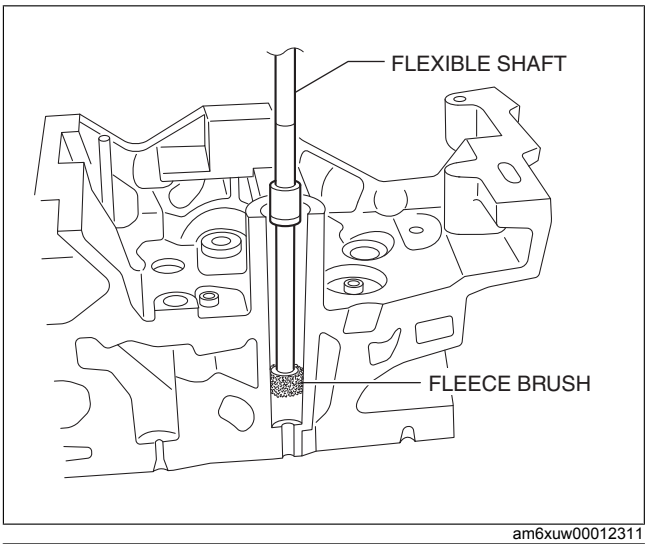
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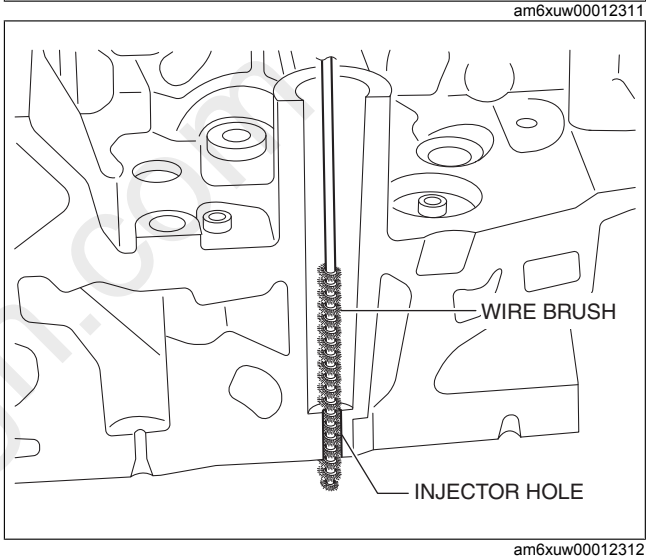
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10. Connect the flexible shaft installed to the fleece brush to the drill, and clean the bottom surface.
11. Clean the carbon that was cut off using an air gun.
12. Insert the extension bar and remove the blanking plug.



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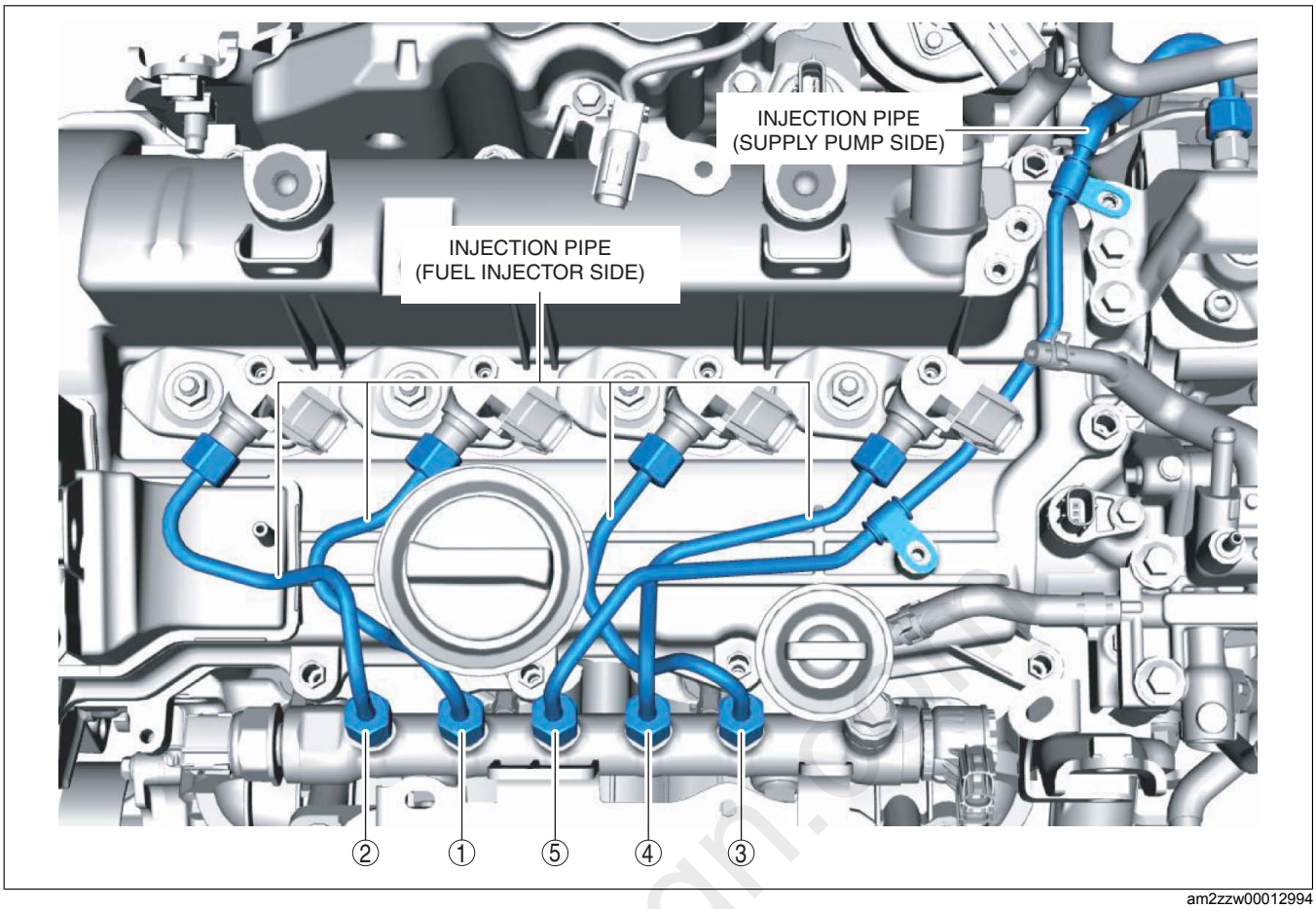
13. Clean the injector hole using a wire brush.
14. Temporarily install the fuel injector and the fuel injector bracket.
15. Temporarily tighten the common rail installation nut.



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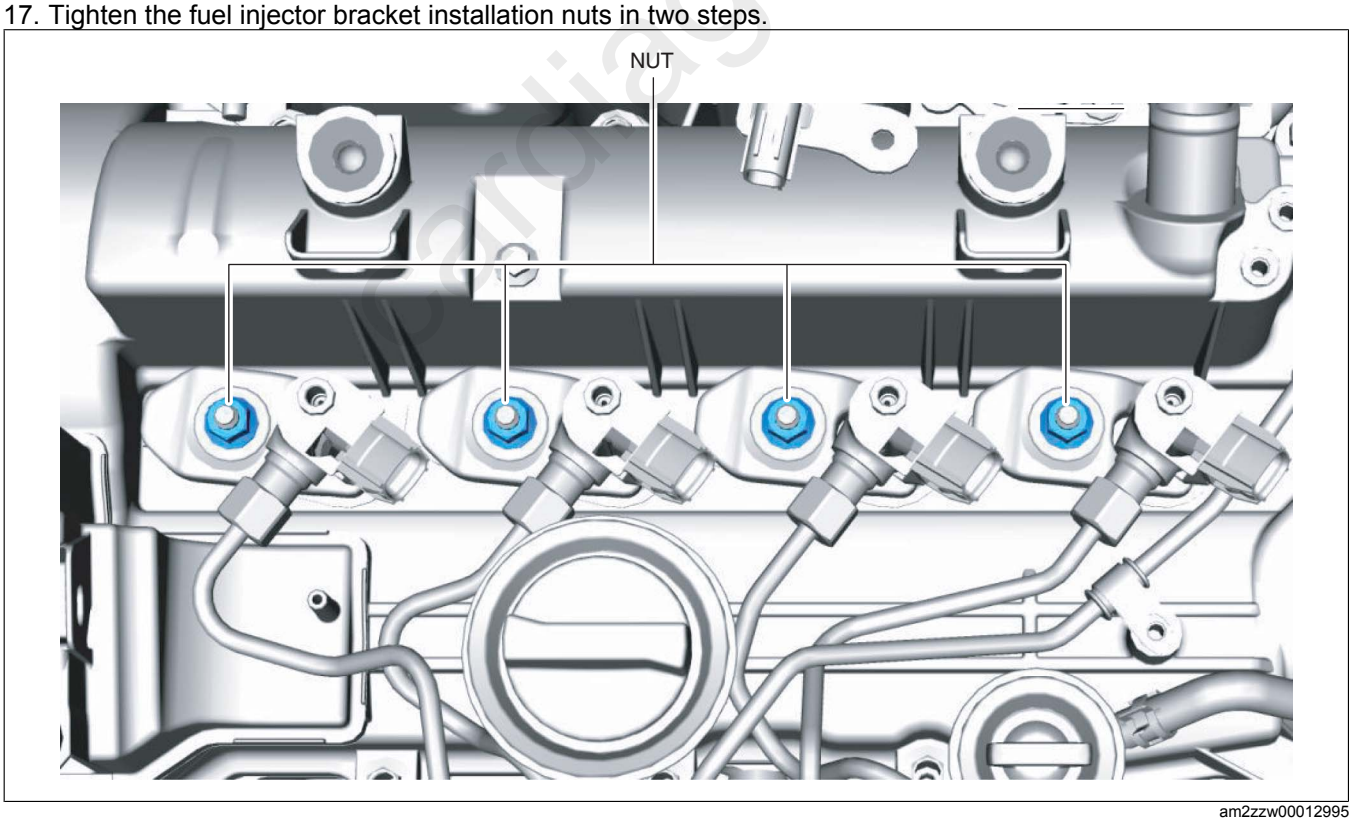
16. Temporarily tighten both ends of the new injection pipe (fuel injector side) and injection pipe (supply pump side) in the order shown in the figure.

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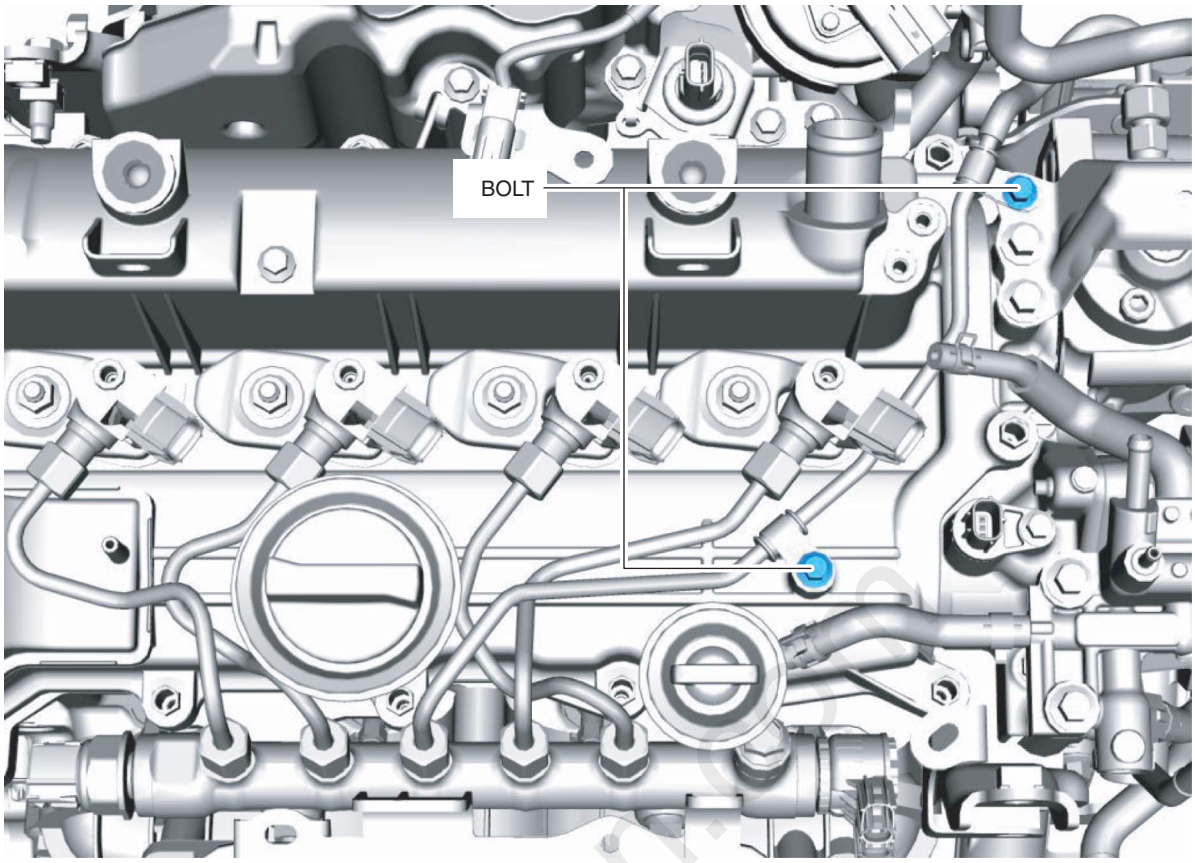
17. Tighten the fuel injector bracket installation nuts in two steps.



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Tightening torque
First pass : 1.5—2.5 N·m {16—25 kgf·cm, 14—22 in·lbf}
Second pass : 117°— 123°

18. Tighten the injection pipe (supply pump side) bracket installation bolts.



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Tightening torque
8—11 N·m {82—112 kgf·cm, 72—97 in·lbf}

19. Tighten both ends of the injection pipe (fuel injector side).

Tightening torque
32—42 N·m {3.3—4.2 kgf·m, 24—30 ft·lbf}

20. Tighten both ends of the injection pipe (supply pump side).

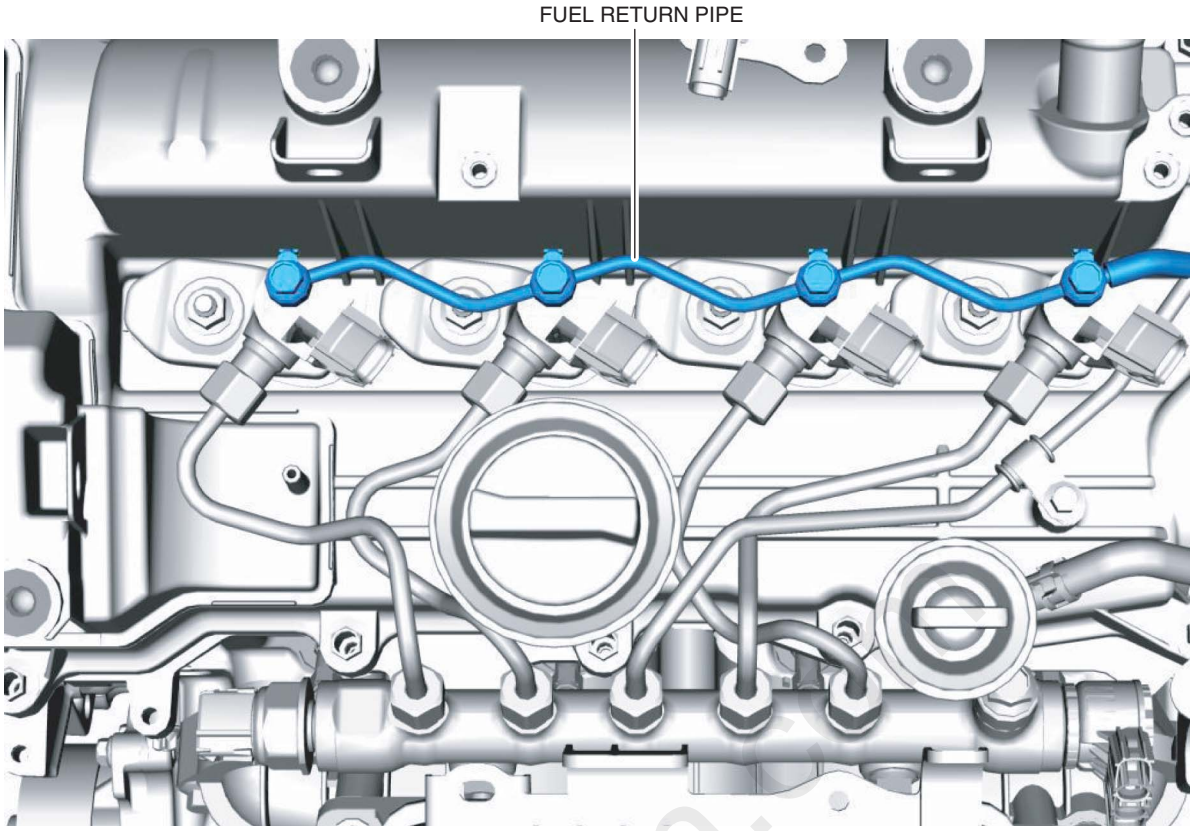
Tightening torque
32—42 N·m {3.3—4.2 kgf·m, 24—30 ft·lbf}

21. Tighten the common rail.

Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}

22. Tighten the fuel return pipe.

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Tightening torque
8—11 N·m {82—112 kgf·cm, 72—97 in·lbf}

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